Worksheet 8-4 to 8-6 A Area Between Curves #1
AP Calculus Name_____

Directions: Sketch the region enclosed by the given curves (you may use DESMOS or a graphing calculator to graph), then find its area. YOU MUST SHOW ALL INTEGRAL WORK.

1.) $y = e^x, y = x^2 - 1, x = -1,\ x = 1$

2.) $y = \sin x\,,\ y = x,\ x = \frac{\pi}{2},\ x = \pi$

3.) $y = x^2 - 4x,\ \ y = 2x$

4.) $x = 1 - y^2,\ \ x = y^2 - 1$

5.) $4x + y^2 = 12,\ \ x = y$

6.) $y = x^2,\ \ y = 4x - x^2$

7.) $y = \cos x\,,\ \ y = 2 - \cos x\,,\ \ 0 \le x \le 2\pi$

8.) $x = 2y^2,\ \ x = 4 + y^2$

9.) $y = \sqrt{x-1},\ \ x - y = 1$

10.) $y = x^3,\ \ y = x$

Answers:

1.) $e - \frac{1}{e} + \frac{4}{3}$ 2.) $\frac{3\pi^2}{8} - 1$ 3.) 36 4.) $\frac{8}{3}$ 5.) $\frac{64}{3}$

6.) $\frac{8}{3}$ 7.) 4π 8.) $\frac{32}{3}$ 9.) $\frac{1}{6}$ 10.) $\frac{1}{4}$